

# WRRRA-M 0.5: Quantized Mass and Continuous Gravity in a Finite Twist Model

Finite calculations of prephenotypic stress in a universe with expansion and twist

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WRRRA-M 0.5 | 1 October 2026

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## Verification input through falsification conditions

| Stage                         | Content of this calculation                                                                                                                                            |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Verification input            | Mass modes in Minimal Computation Cosmology 2.3.2, the carrier conditions of WRRRA-M 0.4, and constitutive responses and calibrations from the individual twist papers |
| WRRRA specific transformation | Separate discrete mass output from continuous geometric output and use a fixed stress response to calculate additional gravity, effective mass, and light deflection   |
| Output                        | Mass modes, 16-channel transport response, filter gaps, finite spherical rotation and lens deflection, and reproduction of the earlier Milky Way calculation           |
| Falsification conditions      | Failed carrier conditions, failed selection gaps, static instability of the stress constitutive law, or inconsistent motion and lensing in the same geometry           |

## Research decision and results

This version selects the B\_C branch: mass can be quantized because it is a phenotype, while gravity itself precedes the phenotype and has no fundamental quantum degrees of freedom. The calculation evaluates measurable gravitational outputs, including the metric and orbits, without introducing an independent Hilbert space for gravity or gravitons. This choice is a model premise of version 0.5. Discrete mass modes are calculated in the specified internal model. The work does not classify the universal impossibility of gravity quantization as an experimental result or a general theorem.

The overall cosmological architecture contains expansion and twist simultaneously. Version 0.5 completes the local twist-stress calculation first. The cause of expansion and global time evolution are not established by this static calculation. Connecting the global and local effects of one stress remains a subsequent task.

The calculation produces three results. First, the earlier calibrations reproduce the transition acceleration of  $1.191812669 \times 10^{-10} \text{ m/s}^2$ . Second, the same constitutive response gives finite spherical rotation and conditional lens deflection, and reproduces the earlier Milky Way model with an RMS of 2.41784 km/s against the linearized reference over 8–25 kpc. Third, both filter gaps vanish for the simplest symmetric 16-channel carrier. This last result concerns the microscopic selection of the target filter: common transport alone does not determine that selection in this prototype.

## Structure inherited from the individual papers

The calculation layers of the individual originals were examined in addition to the integrated account. The twist-stress cosmology paper v1.0 supplies background-stress normalization, the local constitutive response, effective mass, and conditional lensing. The dual-component gravity model v0.4 supplies the decomposition into direct and stress gravity, the denominator behind the 15.7% and 84.3% shares, and the prohibition of double counting across the two interpretations. The galactic-disk-plane paper v1.0 actually adopts the calibrated response across the full acceleration range in its Milky Way calculation and distinguishes an unoriented plane from angular momentum. The finite boundaryless universe paper v1.0 states that expansion and twist coexist and that dynamics must connect global information to local stress. Section 5.3.3 of Causal Horizons v1.1 explicitly identifies B\_C as fundamentally nonquantum gravity.

Section 7.4 of Minimal Computation Cosmology 2.3.2 itself states that fundamental gravity quantization is not included in its proof. Version 0.5 selects B\_C as a research branch without retrospectively changing that statement. The conditional achievements of the original and the stronger model choice made here are therefore preserved separately.

## Distinct status of mass modes and gravity

Periodic boundary conditions in a closed internal model permit integer modes  $n$ . In natural units, the mass relation is

$$n \in \mathbb{Z}, m_n^2 = m_0^2 + \frac{n^2}{R^2}. \quad (1)$$

The reproduction example uses  $m_0 = 0$  and the mass-energy unit  $\mu_M = m_{ec^2}/23$ . This gives  $\mu_M = 22,217.345682$  eV, while  $n = 23$  reproduces the input electron mass-energy of 510,998.95069 eV. This is a reproduction of the internal discreteness and calibration. It does not assert that the physical pole masses of all particles are integer multiples of one common unit. Yukawa coupling, self-energy, and strong-interaction transformations of observed mass retain their downstream roles in version 2.3.2.

Gravity is a geometric response to the total source and boundary state. Every matter channel shares the same geometry. A gravitational output can jump when the source changes discretely without gravity itself acquiring fundamental quantum degrees of freedom. A simple example for the direct component is

$$\Delta g_M(r) = \frac{G \mu_M}{r^2}, \Delta g_M(2r) = \frac{\Delta g_M(r)}{4}. \quad (2)$$

In equation (2),  $\mu_M$  is converted to a mass unit. Even for the same mass-mode change, the output spacing depends on radius and boundary conditions. A discrete source alone does not define a universal gravitational quantum. A finite numerical grid or finite computational precision also does not establish physical quantization of gravity. A complete update law for matter in quantum superposition coupled to classical geometry is outside this static model.

## Coexistence of expansion and twist and the component fractions

The coexistence architecture is recorded by separating the variables of the spatial metric:

$$\gamma_{ij}(t) = a(t)^2 \hat{\gamma}_{ij}[\Theta(t)], H = \frac{\dot{a}}{a}. \quad (3)$$

Here  $a$  describes the isotropic change of scale, while  $\Theta$  describes the twist state. Equation (3) separates variables; it does not derive their dynamics. The present static calculation inside a galaxy fixes the current  $a$  and global boundary state and evaluates the local response. Expansion and accelerated expansion are distinct observables and are not merged into a single causal claim.

The approximately 5% phenotype fraction and approximately 95% remaining before phenotype are editable energy-weighted calibrations. They are not measured fractions of information bits or channel counts. With the present inputs  $f_\Phi = 0.0493$  and clustering-stress fraction  $f_T = 0.265$ , the entire remaining fraction is 0.9507 and the residual background account is 0.6857. That background account is not identified as a microscopic derivation of a separate dark-energy cause. The executed calculation focuses on the local additional-attraction sector represented by  $f_T$ .

Comparing direct matter with clustering stress alone gives  $f_T/f_\Phi = 5.37525$  and a direct-component share of  $f_\Phi/(f_\Phi + f_T) = 15.6857\%$ . The approximately 5% and 15.7% figures can coexist because they use different denominators. This cosmic mean source ratio is not imposed as the gravitational ratio at every galactic radius.

## Information before phenotype and a closed universe

The upstream hypothesis is fixed as follows. The physical load of information that has not been rendered as a phenotype determines twist. Global gluing preserves that twist and constructs a finite universe without a boundary. This requires a transformation from information content to physical load. At the present stage,  $E_{pre}$  assigns energy weights to the information states:

$$E_{pre} = \sum_a w_a I_a, u_{pre} = \frac{E_{pre}}{V}, \zeta \kappa_{tot}^2 = \frac{16\pi G}{c^4} \frac{E_{pre}}{V}. \quad (4)$$

The model weights  $w_a$  convert information states into an energy load. Once this transformation is fixed, the remaining information content is an input to the twist-rate calculation. At fixed volume and stiffness,  $\kappa_{tot}^2$  is therefore proportional to  $E_{pre}$ . No universal rest mass is assigned to an arbitrary information bit.

A three-torus  $T^3$ , periodically glued in three directions, provides one constructibility model of the global identification. This model treats all local patches of one common Source equally. If  $\Theta_i = \kappa_i L$  records twist along the three closed curves of representative static length  $L$ , then

$$V = L^3, Q = \sum_{i=1}^3 \Theta_i^2 > 0, E_{pre} = \frac{\zeta c^4 Q}{16\pi G} L, L = \frac{16\pi G E_{pre}}{\zeta c^4 Q}. \quad (5)$$

For finite  $E_{pre}$ , positive stiffness, and fixed nonzero gluing records,  $L$  and the volume are finite, and  $T^3$  has no boundary. Equation (5) is an exact algebraic relation in the chosen gluing model. It provides one possible implementation from information load through twist to closed geometry. It is not a theorem that every twist automatically selects  $T^3$ , or a measurement of the actual cosmic topology. Changes in  $L$  over time can be linked to the expansion variable, but the evolution equation is not calculated here.

The model does not impose the unconditional rule that more information must always mean more twist. At fixed volume, the twist rate increases. If the gluing records are fixed while the volume also changes, the length changes according to equation (5). Because  $\zeta$ ,  $\Theta_i$ , and the information weights remain unspecified, the actual cosmic length is not fixed to one number.

Separating the local clustering-stress fraction  $f_T$  from the entire remaining load  $f_{pre}$  preserves the local scale of the earlier individual papers while admitting the new upstream hypothesis. Under the additional assumption that the same quadratic load map extends to the whole hidden sector, the present calibration  $f_{pre} = 0.9507$  gives the normalized index  $\sqrt{6f_{pre}} = 2.388347$ . This is an index of total load rather than a measured angle. The local additional-attraction scale is still calculated with  $f_T = 0.265$ .

## Twist normalization in the present universe

The quadratic stress energy and the closure relation across scales are taken from the earlier individual paper:

$$u_T = \frac{\zeta c^4 \kappa^2}{16 \pi G}, \rho_c = \frac{3 H_0^2}{8 \pi G}, f_T = \frac{u_T}{\rho_c c^2}. \quad (6)$$

$$a_T = c H_0 \sqrt{\frac{f_T}{8}}, \zeta \kappa^2 = 6 f_T \left( \frac{H_0}{c} \right)^2. \quad (7)$$

The inputs are  $c = 299,792,458$  m/s,  $G = 6.67430 \times 10^{-11}$  in SI units,  $H_0 = 67.4$  km/s/Mpc, and  $f_T = 0.265$ . The value of  $c$  is a defined input for unit conversion and is not a prediction target. The factor  $1/8$  and the matched-response connection in equation (7) are previously calibrated model elements. Once known values fix the model, the quantities calculated by this relation remain WRRA outputs.

The outputs are  $a_T = 1.191812669 \times 10^{-10}$  m/s<sup>2</sup>,  $\rho_c = 8.532855164 \times 10^{-27}$  kg/m<sup>3</sup>, and  $\zeta \kappa^2 = 8.440621409 \times 10^{-53}$  m<sup>-2</sup>. The normalized index  $\sqrt{6f_T} = 1.260952$  is not a directly measured twist angle. The stiffness  $\zeta$  and twist rate  $\kappa$  are not separately determined.

## Calibrated constitutive response and static action

The constitutive response adopted in the disk-plane paper is used unchanged across the full acceleration range:

$$y = \frac{g_M}{a_T}, v(y) = \frac{1}{1 - e^{-\sqrt{y}}}, g = v(y) g_M, g_T = g - g_M. \quad (8)$$

The response function is an observationally calibrated input, not a uniquely derived function of the upstream common carrier. This version actually evaluates it within the internal calculation. Effective stress interpreted as a source and modified geometry are not added twice in one calculation.

The spherical static geometric response of  $B_C$  admits the following constructed action. Define  $x = g/a_T = yv(y)$  and  $\mu(x) = y/x$ :

$$E[\Phi] = \int d^3 x \left[ \frac{a_T^2}{8 \pi G} F(Y) + \rho_b \Phi \right], Y = \frac{|\nabla \Phi|^2}{a_T^2}, F_Y = \mu. \quad (9)$$

$$\nabla \cdot \left[ \mu \left( |\nabla \Phi| / a_T \right) \nabla \Phi \right] = 4 \pi G \rho_b. \quad (10)$$

This static variational structure uses the established mathematics of AQUAL. The function  $\mu$  is reconstructed to produce the same spherical response as the selected  $v$ . Outside spherical symmetry, QUMOND and AQUAL are not treated as the same equation. Equation (9) does not complete a four-dimensional covariant action.

$$F[Y(y)] = \int_0^y 2q \frac{dx}{dq} dq, \frac{dx}{dy} = \frac{1 - e^{-s} - (s/2)e^{-s}}{(1 - e^{-s})^2} > 0, s = \sqrt{y} > 0. \quad (11)$$

Positivity follows from  $e^s > 1 + s/2$ . The transverse eigenvalue  $\mu$  and longitudinal eigenvalue  $d(\mu x)/dx = dy/dx$  of the static variation are positive for  $x > 0$ . Numerical differentiation in the code reproduces  $dF/dY = \mu$  with a maximum relative error below  $4.11 \times 10^{-10}$ . At  $g = 0$  the equation degenerates, so the strict ellipticity check excludes the center. This result is not extended to a proof of stability or causality of the entire covariant model.

## Finite spherical calculation and the dark mass account

The spherical test source has a total baryonic mass of  $6 \times 10^{10}$  solar masses and Plummer scale  $b = 3$  kpc. This reproducibility input is distinct from the actual Milky Way model. The declared calculation patch has radius 200 kpc.

$$M_b(<r) = M \frac{r^3}{(r^2 + b^2)^{3/2}}, g_M = \frac{G M_b(<r)}{r^2}, v_c^2 = r g. \quad (12)$$

$$M_{T,eff}(<r) = \frac{r^2 g_T}{G}, \rho_{T,eff} = \frac{1}{4\pi r^2} \frac{dM_{T,eff}}{dr}. \quad (13)$$

Equation (13) is an account that expresses the additional effect of the same geometry as a Newtonian-equivalent mass. It does not count quantized dark particles or represent matter rest mass. The local microscopic twist rate  $\kappa(r)$  is not reported as though it had been directly measured through equation (13).

| Radius<br>kpc | Direct speed km/s | Total speed km/s | Additional gravity / direct<br>gravity | Effective stress mass solar<br>masses |
|---------------|-------------------|------------------|----------------------------------------|---------------------------------------|
| 1             | 90.34             | 102.65           | 0.291                                  | 5.523e+08                             |
| 3             | 174.39            | 193.78           | 0.235                                  | 4.979e+09                             |
| 8.2           | 161.45            | 207.51           | 0.652                                  | 3.240e+10                             |
| 20            | 111.71            | 192.26           | 1.962                                  | 1.139e+11                             |
| 50            | 71.65             | 182.65           | 5.499                                  | 3.282e+11                             |
| 100           | 50.77             | 179.14           | 11.453                                 | 6.862e+11                             |

At low acceleration,  $g \approx \sqrt{a_T g_M}$  and  $v_f^4 = G M a_T$ . The deep-regime speed scale of the test source is 175.5177 km/s. The slightly higher speed at 100 kpc follows from the fixed transition response. Effective stress density remains nonnegative throughout the 0.01–200 kpc patch, and the maximum relative error in spherical flux conservation is  $3.42 \times 10^{-16}$ . Global stress outside the patch and the gluing conditions of the finite universe are not specified in this calculation.

## Conditional lensing in the same geometry

The condition  $\Phi = \Psi$  from the individual stress paper is used to calculate light deflection from the same potential. This condition must be checked in a covariant completion.

$$\hat{\alpha}_R(b) = \frac{2}{c^2} \int_{-\sqrt{R^2-b^2}}^{\sqrt{R^2-b^2}} g(\sqrt{b^2+z^2}) \frac{b}{\sqrt{b^2+z^2}} dz. \quad (14)$$

| Impact radius kpc | Total deflection arcsec | Direct component rad | Twist component rad |
|-------------------|-------------------------|----------------------|---------------------|
| 5                 | 0.5928                  | 1.6883e-06           | 1.1855e-06          |
| 10                | 0.5356                  | 1.0523e-06           | 1.5443e-06          |
| 20                | 0.4730                  | 5.5875e-07           | 1.7342e-06          |
| 50                | 0.3955                  | 2.2159e-07           | 1.6960e-06          |

The table gives the physical deflection accumulated inside the 200 kpc patch. It excludes the geometry outside the patch, observer-lens-source distance factors, and the observed image angle. No new stress dedicated to lensing or separate normalization is introduced. The conditional result is therefore that the same static response produces both motion and light output. Fitting observed lenses and covariantly deriving gravitational slip remain separate tasks.

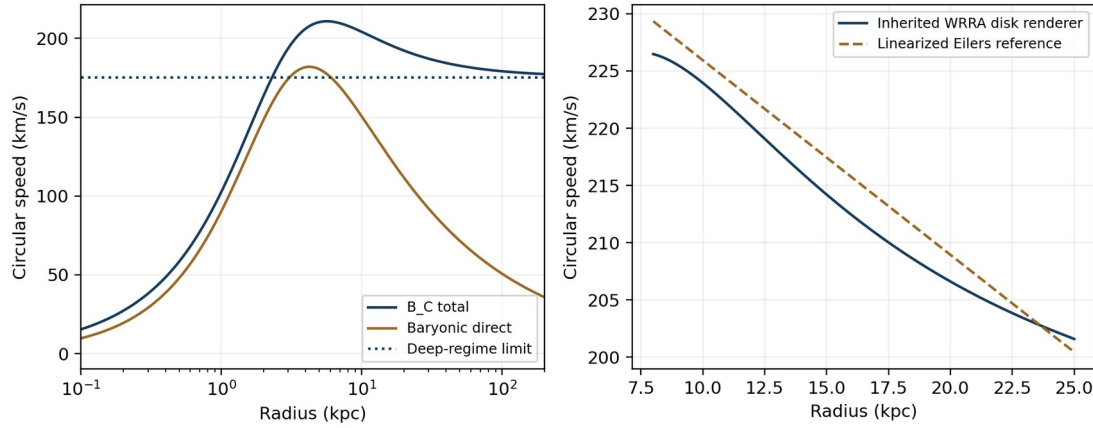
## Reproduction of the individual Milky Way calculation

The adopted inputs are unchanged: stellar disks of  $35.66 \times 10^9$  and  $11.25 \times 10^9$  solar masses with scales of 2.53 and 3.38 kpc; HI and H<sub>2</sub> disks of  $11.0 \times 10^9$  and  $1.2 \times 10^9$  solar masses with scales of 7.0 and 1.5 kpc; and a bulge of  $9.13 \times 10^9$  solar masses with Hernquist scale 0.70 kpc. The modified-Bessel-function contributions of the exponential disks and the direct squared speed of the bulge are summed before equation (8) is applied.

| Radius kpc | Direct speed km/s | WRA speed km/s | Additional gravity / direct gravity |
|------------|-------------------|----------------|-------------------------------------|
| 5          | 192.44            | 221.03         | 0.319                               |
| 8.2        | 182.55            | 226.34         | 0.537                               |
| 12         | 161.05            | 220.09         | 0.868                               |
| 16.5       | 138.93            | 211.60         | 1.320                               |
| 20         | 125.92            | 206.61         | 1.692                               |
| 25         | 112.05            | 201.59         | 2.237                               |

Over 8–25 kpc, the RMS against the linearized Eilers reference of  $229 - 1.7(R - 8.2)$  km/s is 2.41784 km/s, reproducing the earlier paper's 2.42 km/s. This comparison is not a likelihood using individual observational data

points and covariance. The disk calculation also retains the earlier algebraic Renderer approximation. It is not classified as a three-dimensional disk solution of the present spherical variational equation.



### Rotation speeds of the spherical test model and the inherited Milky Way Renderer

The unoriented plane is inherited as  $P = I - nn^T$ . Replacing  $n$  with  $-n$  leaves  $P$  unchanged, and  $P^2 = P$ . The code gives a sign-invariance defect of zero and a projection defect below  $5.56 \times 10^{-17}$ . This plane field does not define the sense of rotation or angular momentum. Because the directional stress  $q_T$  is not fixed, vertical thickness and restoring force are not outputs of this version.

### Executed carrier test and the selection boundary

The next calculation specified by 0.4 is executed for the simplest homogeneous carrier. Sixteen channels share the same principal transport structure and periodic internal model. The internal cycle-lattice Laplacian has eigenvalues  $4\sin^2(\pi j/N)$ . For  $N = 128$ , the positive Green-function intensities at five declared frequencies are averaged to extract each channel's scalar signature. Channel-specific lower-order operators are identical in this test.

With orthogonal channel injections into the same carrier, the transport Gram operator is a positive multiple of  $I_{16}$ . After normalization, its rank is 16, its smallest eigenvalue is 1, and the squared transport norm of the neutral channel is also 1. However, the eight gauge-invariant scalar responses are identical. The squared-mismatch score of 0.4 therefore gives

$$\Delta_L = 0, \Delta_R = 0, S(F_{DX}) = S(F_{XX}) = S(F_{DD}) = S(F_{XD}) = 0. \quad (15)$$

The homogeneous test passes the transport conditions but does not uniquely select the target filter. This is an exact negative result for the fixed prototype, not an impossibility theorem for all of WRR-M. The next selection calculation requires differences in lower-order responses actually generated by channel origin and residue. Full transport rank and positive selection gaps are distinct results. The gauge-neutral sixteenth channel is also not automatically identified with the prephenotypic stress sector.

## Changing calibrations and falsification conditions

The component fractions are editable inputs in parameters.json. Example changes give the following outputs:

| Clustering stress fraction | Matter fraction | Entire remaining fraction | Transition acceleration m/s <sup>2</sup> | Spherical limiting speed km/s |
|----------------------------|-----------------|---------------------------|------------------------------------------|-------------------------------|
| 0.200                      | 0.0493          | 0.9507                    | 1.0354e-10                               | 169.45                        |
| 0.300                      | 0.0493          | 0.9507                    | 1.2681e-10                               | 178.26                        |
| 0.265                      | 0.0800          | 0.9200                    | 1.1918e-10                               | 175.52                        |

Changing the clustering-stress fraction  $f_T$  changes both  $a_T$  and the speed scale. Changing only  $f_\Phi$  changes the total hidden fraction and mean source ratio, while the local acceleration scale remains unchanged at fixed  $f_T$ . The approximately 5% and 95% fractions are therefore not made structurally immutable.

The calculation must be rejected or revised under the following conditions. If Gram positivity or full transport rank fails for the declared carrier, the transport branch fails. If positive selection is asserted but either actual gap is nonpositive, that selection claim fails. Negative static constitutive eigenvalues or failed spherical conservation invalidate the static implementation. If the same geometry cannot account for both motion and lensing, the present slip condition or stress law must be revised. If global dynamics cannot support simultaneous expansion and twist or total conservation, that subsequent completion fails.

Model revision using known observations is permitted. Revised inputs, constitutive laws, and versions must be recorded, and the outputs of the revised model evaluated again. An external theory that has not actually been calculated within WRRR is not called integrated. The executed items of this version are recorded in the code and result account. The covariant action, hybrid updating of matter and classical geometry, global boundary dynamics, and vertical stress remain subsequent tasks.

## Claim account and completion boundary

| Claim                                     | Status                                   | Basis in this version               |
|-------------------------------------------|------------------------------------------|-------------------------------------|
| Discrete internal mass modes              | Exact result in the specified model      | Periodic boundary and integer modes |
| Fundamentally nonquantum gravity          | B_C model premise                        | Selection of this branch            |
| Coexistence of expansion and twist        | Preserved structural assumption          | Equation (3)                        |
| Twist acceleration scale                  | WRRR output after calibration            | Equations (6) and (7)               |
| Static constitutive action and positivity | Conditional construction and calculation | Equations (9) through (11)          |
| Spherical stress mass and rotation speed  | Finite model outputs                     | Equations (12) and (13)             |

| Claim                                                     | Status                             | Basis in this version                                 |
|-----------------------------------------------------------|------------------------------------|-------------------------------------------------------|
| Finite-patch lens deflection                              | Conditional output with no slip    | Equation (14)                                         |
| Earlier Milky Way speeds                                  | Reproduction of a calibrated model | RMS 2.41784 km/s                                      |
| Target selection by the homogeneous carrier               | Not achieved in this test          | Both gaps are zero                                    |
| Microscopic derivation of the stress law from the carrier | Unresolved                         | Lower-order responses and a connecting law are needed |

The completion boundary of 0.5 is to fix the quantization status of  $B_C$ , actually calculate the additional gravity, effective mass, and conditional light effects of twist stress, and disclose the filter-selection result of the simple carrier. The finite calculation reaches that boundary. Structural originality is evaluated through the connection between discrete mass and continuous geometry, matter output and upstream stress, and common transport for motion and light. Microscopic filter selection and full covariant integration are recorded as subsequent calculations.

## References and reproduction material

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In the reproduction package, `compute.py` reads `parameters.json` and calculates mass modes, carrier responses, spherical stress, finite-patch lensing, and Milky Way speeds. The `results.json` file records the inputs, outputs, and numerical checks together. The `source_en.md` file is the mathematical source of this English edition. The calculation dependencies are `numpy`, `scipy`, and `matplotlib`.

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